📄 VSEC10 PS — Docker Compose (Multi-Container AI App)

🔹 1. Install Docker + Compose

Install Docker (Docker Compose comes built-in in Docker Desktop / recent Docker)

Keep Docker Desktop open in bg

Verify:

docker --version
docker compose version

👉 Run in:

- Windows → Git Bash
- Ubuntu → Terminal

🔹 2. Create Project Folder

mkdir ai-compose-project
cd ai-compose-project

🔹 3. Create Folder Structure

mkdir backend frontend
touch docker-compose.yml

🔹 4. Create Backend (AI Model)

cd backend
touch app.py requirements.txt Dockerfile

🔹 Write backend code

Open file

Windows:

notepad app.py

Ubuntu:

nano app.py

Code

```
from flask import Flask

app = Flask(__name__)

@app.route('/')
def home():
    return "AI Model Running!"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

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requirements.txt→

notepad requirements.txt # Windows
nano requirements.txt # Ubuntu

Add:

flask

Dockerfile (backend)→

notepad Dockerfile # Windows
nano Dockerfile # Ubuntu

```
FROM python:3.10
WORKDIR /app
COPY . .
RUN pip install -r requirements.txt
EXPOSE 5000
CMD ["python", "app.py"]
```

- 5. Create Frontend

cd ..
cd frontend
touch index.html Dockerfile

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index.html→

notepad index.html # Windows
nano index.html # Ubuntu

Paste:

<h1>Frontend Connected to AI Backend</h1>

Dockerfile (frontend)

notepad Dockerfile # Windows
nano Dockerfile # Ubuntu

FROM nginx:latest
COPY index.html /usr/share/nginx/html/index.html
EXPOSE 80

- 6. Create docker-compose.yml

Go back to root:

cd ..
notepad docker-compose.yml # Windows
nano docker-compose.yml # Ubuntu

Code:

docker-compose.yml

```
version: '3'

services:
 backend:
  build: ./backend
  ports:
```

```
    - "5000:5000"

  frontend:
    build: ./frontend
    ports:
      - "8080:80"

  database:
    image: mysql:5.7
    environment:
      MYSQL_ROOT_PASSWORD: root
```

🔹 7. Build and Start Containers

docker compose up --build

👉 Run in project root folder

🔹 8. Verify Running Containers

docker ps

👉 You will see 3 containers: frontend,backend,database

🔹 9. Test Application on Browser

Frontend:

http://localhost:8080

Backend:

http://localhost:5000

🔹 10. Stop All Containers

docker compose down

(Check the full containers on Docker Desktop with Build info)